NIHR | National Institute for Health and Care Research

NIHR Doctoral Fellowship (DF)

Funding, Support or Development Opportunity	2025/303 NIHR Doctoral Award (Cohort1)

SUMMARY INFORMATION

Application Title	How do airway dynamics change in children with Spinal Muscular Atrophy during assisted coughing?
Contracting Organisation	GREAT ORMOND STREET HOSPITAL FOR CHILDREN NHS FOUNDATION TRUST
Partner Organisation(s)	UCL Great Ormond Street Institute of Child Health
Start Date	01/04/2026
End Date	31/03/2029
Duration (months)	36
% FTE Commitment	100%
Percentage of award time undertaking practice/clinical work	0%
Total Research Costs Requested	£355,274.76
Has this application been previously submitted to this or any other funding body?	Yes **Grant Reference**: NIHR305197 **Funder**: NIHR **Please Provide Details**: Unsuccessful DCAF application Response to Feedback: Thank you to the previous reviewers for their comments and helpful feedback. I have taken them into account and below is the response. Response to Reviewer 1 Published but not first author recently: Since the previous application I have sent two papers for publication as first author, both of which are currently in press. Systematic review: Comments were focused on the systematic review and the evidence available. This was reviewed and it was deemed more appropriate to complete a scoping review. This has now been completed and is one of the papers in publication. Focus groups: Due to a change in the project design, these have been removed. Exploration of the pattern of breathing: , further discussion has been added as to exploration and reasoning for the use of this work package. The data will provide lung function data, which will be used to assess if there is an improvement in lung volume as a result of optimised cough assist settings. Number of interventions: timing has been added to the proposal to give an idea of time frame. Host Statement: Adapted to demonstrate support beyond the fellowship Timing: The project has been adapted, and the complexity has been

	reduced. This is in response to concerns over the time frame of the work packages. With the adjustment in the overall project plan and prior completion of the scoping review, the data collection has been extended to 9 months. More information was requested regarding the pilot data. On reflection, this was deemed more of a useability review and additional information has been provided. Response to Reviewer 2 Building a clinical academic career: The proposal gives increased information regarding future plans. The Question: It was highlighted the original question was not a question and this has been adapted to reflect the project. Qualitative study design: The project has been adapted, and the complexity has been reduced. This is in response to concerns over the time frame of the work packages. As a result, the qualitative research has been removed, with a new focus on tolerance of the new tests. The qualitative work looking at a questionnaire and patient reported outcome measure is hoped to be the completed in the future as part of a post-doc. Risks and considerations: These have been addressed in each work packages Fatigue for patients: The tests will be planned with regular breaks and options will be given to families to allow for flexible. The tests are not overly long and this will help reduce fatigue. Supervisors and clinical development: The supervisory team has been reviewed and strengthened with regards to overall previously supervised students to completion. A methodologist and statistician have also been added to the team. The international collaborations have also been reviewed with the addition of work with the Italian network with regards to the OEP. Response to Reviewer 3 Experience and Exposure of applicant: Since the last application, I have completed 2 papers which are in press. I have also become PI on a study for which I have completed all stages of the project, including research and development procedures, ethical approval, study set up and recruitment. Systematic review: Comments were focused on the systematic review and the evidence available. This was reviewed and it was deemed more appropriate to complete a scoping review. This has now been completed and is one of the papers in publication. Focus on SMA: It was also felt the project was niche to the area of SMA. It has been highlighted through the impact of how this project can be translated into other illness with far reaching impact. Physiotherapy dissemination: The reviewers also highlighted the importance of physiotherapy-based conferences/dissemination, and as a result JIVD has been added which is a conference focused on the multidisciplinary team and has a specific area related to respiratory physiotherapy.
Partnership Fellowship	Not Applicable

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LEAD APPLICANT & RESEARCH TEAM

Lead Applicant

Full Name	Mrs Lisa Edel
Organisation	GREAT ORMOND STREET HOSPITAL FOR CHILDREN NHS FOUNDATION TRUST
Department	Physiotherapy
Professional Background	Allied Health Professional
Detailed Professional Background	Physiotherapist
Professional Registration	HCPC: PH88636
ORCID	0000-0002-9352-5388
Research Degree Status	Not registered
Date of Registration	
Percentage FTE uptake of research degree	
Date of submission of research degree	
Date of award of research degree	
Contextual Factors	Not applicable

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APPLICATION DETAILS

Scientific Abstract

Research Question
How do airway dynamics change in children with Spinal Muscular Atrophy during assisted coughing?
Background
Spinal Muscular Atrophy(SMA) is a life-limiting condition affecting children's ability to move limbs, swallow, breathe and cough effectively, impacting quality of life and life expectancy. Patients develop increased secretions, causing recurrent chest infections and hospital admissions. Physiotherapy relies on devices called cough-assists (Mechanical Insufflation Exsufflation-MIE) to help prevent infections. MIE delivers positive pressure promoting deep inhalation followed by a rapid switch to negative pressure to mimic a cough. MIE are programmed on specific settings for children to use at home. However, there is limited understanding of what occurs in the child's airway when breathing, coughing, and using MIE. The main concern is potential airway collapse and MIE being ineffective. Understanding the upper airway response to using MIE is vital for improved patient outcomes.
Aims
Investigate reasons for variability in effectiveness of MIE in children, establishing optimal settings for MIE in patients with SMA
Assess variations in lung function with optimised MIE settings
Evaluate tolerance and acceptance of new assessment techniques.
Objectives
Work Package(WP)1a: Assess the upper airway dynamics using Transnasal Fibreoptic Laryngoscope(TFL) in children with SMA
WP1b: Evaluate/optimise upper airway response patterns to MIE
WP2: Assess changes in respiratory patterns/lung volumes utilising opto-electronic plethysmography(OEP) with baseline and optimised MIE settings
WP3: Measure patients/carers tolerance and acceptability of assessment techniques
Method
An interventional quantitative study consisting of three WPs, supported by community steering groups. 45 patients will be recruited. In WP1 Patients will be assessed with TFL during normal breathing and use of home MIE settings. Airway status will then be analysed with MIE being adjusted for optimal airway status. In WP2, OEP will measure differences in breathing patterns and lung volumes, before MIE, at baseline MIE settings and on adjusted MIE settings. During WP1 and WP2, patients/carers will rate tests in terms of tolerance and acceptance on a visual scale (WP3). Statistical analysis will be conducted. Data synthesis will confirm challenges, strengths, and areas for improvement.
Timelines for delivery
WP1, 2 and 3: 9-18Months
Knowledge mobilisation, dissemination, and impact
Improved secretion management will reduce hospitalisations improving health and wellbeing. The study provides insight into airway dynamics and will translate into other paediatric conditions. Results will be shared via podcasts and meetings with parent support groups/charities, and presentations at meetings/conferences, throughout the UK and internationally, including submission to appropriate peer reviewed journals.
Research Inclusion
The study focuses on paediatric patients with SMA, which as a rare disease serves an under-represented population from all cultural backgrounds. Adjustments have been made to ensure patients with behavioural and communicative concerns are able to participate. Patient information sheets will be available to all and adapted for different communication styles. An animated video has been created to aid inclusion. Characteristics related to inclusion of participants will be reported. Ongoing reviews will ensure equality, diversity and inclusion are considered.

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Plain English Summary

Spinal Muscular Atrophy (SMA) is a condition which affects how long you can live, mostly affecting children, which weakens muscles, including those needed to move limbs, swallow, breathe and cough. As a result of weakened breathing muscles, a thick and sticky substance known as mucus builds up in the child's lungs, causing difficulty breathing. This leads to infections in their lungs and repeated hospital visits affecting children's quality of life and life expectancy.

Physiotherapists work with these children and their parents to prevent them from becoming unwell. One way these children can be helped is by daily use of a machine referred to as a cough-assist, which mimics a cough clearing mucus from the lungs. It requires a child to temporarily wear a mask over their nose and mouth as air is blown into their lungs to help them take a big breath in, followed by creating a vacuum to move mucus out.

In SMA, we need to understand what happens to the child's airway (breathing tube) whilst breathing, during coughing and when using cough-assist.

We need more knowledge about how cough-assist settings help improve mucus clearance, reducing the risk of chest infections, improving quality of life for patients. This would also reduce hospital visits and costs to the NHS.

Research aims and design:

This will be based at Great Ormond Street Hospital.

There are three parts.

1. Using a small (2mm) camera through the nose, we will view and assess what happens to the airway of children with SMA when:

a. they breathe independently and use their home cough-assist

b. we change cough-assist settings to see if this improves coughing.

These tests will not hurt. We will record what the camera shows us.

2. Measuring the breathing patterns of children when relaxed and when using cough-assist.

Patients will lie relaxed on a bed, breathing normally. Between 24-52 temporary markers will be put on their chest. The number of markers depends on the size of the patient. Eight cameras in the ceiling will pick up signals from the markers and change the information into the child's breathing pattern. We will be able to measure the amount of air in the patient's lungs, breath by breath. We will measure breathing at rest, breaths with the original cough-assist settings as well as breaths with adjusted cough-assist settings from part 1. We will then measure the differences in breathing patterns. This will highlight any changes in the way children with SMA breath, having potential benefit in clearing more mucus.

3. Understand patients', parents' and carers' views, tolerance, and acceptance of the new assessment techniques.

During the tests, patients will be asked to rate the new tests in terms of acceptance, comfort, and tolerance on a visual scale.

Sharing results

We will include patients and their parents during the study. They will guide us in the best ways to share information about what we find. We will also share what we find at meetings and conferences (In the United Kingdom and internationally). Papers will be shared through medical and physiotherapy journals.

Working with People and Communities Summary

This proposal was developed in collaboration with parents, carers and patients of Great Ormond Street Hospital (GOSH), charity partners and medical colleagues from across the United Kingdom who are linked to SMA. Project proposal/relevance, study design and concerns were shared.

The medical community stressed the important link between airway dynamics and lung volumes, and monitoring of patient tolerance to new tests. They highlighted this work as visionary and a way to anticipate impact of a changing phenotypes of SMA

A patient and communities' group of 6 parents, 3 patients and 2 charity partners, aged 16-42 from diverse socio-economic and ethnic backgrounds, were consulted on the plain English summary and joined focused discussions. The Child and Young Person Group at GOSH contributed to the original planning phase and plain English summary.

The families confirmed importance of the research, highlighting it would help children stay out of hospital, and their feedback has informed proposal development and design. Families reported it 'gives

us a door to open up and see what is happening inside' and 'we are grateful you are looking into the respiratory side of SMA'.
Some parents raised concerns, and we revised/added activities to accommodate this feedback:
Transnasal fibreoptic laryngoscope (TFL): Concerns were regarding appearance, feel and what happens if the child dislikes it. A child friendly video/cartoon explaining the procedure was suggested to help with understanding. This has since been created. Patients requested to see the new equipment and all equipment detailed in the study will be accessible to all patients.
Fatigue: The community requested testing for the TFL be completed in the morning, followed by the opto-electronic plethysmography in the afternoon. We have ensured families will be able to take regular breaks. The data collection period was made longer to allow for more spacing of patients to enable more time per test.
Study Feedback: Regular feedback on the study was requested, ensuring it was appropriate and all-inclusive. Coffee mornings/online meetings on study progression are planned.
Managing the Research
A communities' group will be created, made up of 3 young adults living with SMA, 4 parents of children with SMA and 3 representatives from charity partners. Recruitment will be through email correspondence and social media, with a follow up meeting to discuss role. Face to face and online meetings were requested. There will be 3 yearly face to face meetings and online groups 5-6 monthly. Discussions via emails/telephone calls will occur at key project timepoints.
Parents/Patients/Charity Partners will be offered to:
• Participate in training for involvement in advisory groups, provided by Imperial College
• Guide the project throughout the research process
• Explore ethical issues
• Assist with plain English summaries/updates
• Provide patient representation, parent perspective and SMA network links.

International and national medical professionals will form a steering group, focusing on ethical issues/study design/time management/concerns arising throughout the project.
Dissemination
Steering groups will assist with writing final reports/summaries of results and will be acknowledge for their involvement. Their role in facilitating publication of findings on charity/SMA specific websites and dissemination through their informal networks will be invaluable to ensuring dissemination to all audiences.

SUPPORT AND MENTORSHIP

Full Name	Dr Tiina Maarit Andersen
Organisation	Western Norway University of Applied Sciences
Email Address	tiina.maarit.andersen@helse-bergen.no
Role	Practice Supervisor
Description of Support Provided	I have know Lisa since 2021 through her impressive contributions to the field of paediatric respiratory physiotherapy, particularly her work with children affected by neuromuscular conditions. Her innovative approach to clinical practice and research in this niche area has gained international recognition and she has significantly contributed to improving patient care. Lisa has a robust research foundation prior to her PhD work, having participated in large-scale clinical trials and the development of novel respiratory outcome measures that are widely used internationally. Her dedication to translational research is evidenced by the implementation

of her respiratory assessment tool in clinical settings worldwide, particularly for children with Spinal Muscular Atrophy. Her academic achievements, such as receiving the Lea Rose Award, further underscore her research capabilities. Additionally, her extensive international collaborations reflect her commitment to advancing standards of care and her ability to engage effectively with a global research community.

Lisa has several supervisors, where most are located in the UK, where Lisa is collecting the data for the PhD. She has access to the patient population described in the project plan, and the equipment to perform the examinations. Additionally, I will contribute for Lisa to visit us and learn the method of continuous laryngoscopy during MI-E manoeuvres from the research group that is currently world leader on this.

I will support Lisa with the method of continuous laryngoscopy during the MI-E interventions, how to plan the examinations, perform them, and how to interpret the data, which is a method I developed in 2010. I developed the idea to use laryngoscopy to study the upper airways during dynamic physiotherapy treatment with a cough assist device. My PhD involved assessing the feasibility of the examination and determining normal findings in healthy individuals. Subsequently, I examined all Amyotrophic Lateral Sclerosis patients in our region and identified abnormalities in these complex cases. This was followed up to five years throughout disease progression, describing how these changes evolve over time. This research has significantly improved the clinical treatment of the most vulnerable and challenging patients with chronic respiratory failure. Following my PhD (2018), I continued working as a postdoctoral researcher with a 50% grant in 5 years from Western Norway Health Authorities, focusing on enhancing clinical respiratory support treatment for ALS patients. This multicentre study includes participants from Bergen, Norway, and Porto, Portugal. Additionally, we are investigating the role of the larynx in another motor neuron disease affecting both children and adults. Lisa project aligns with my ongoing areas of research.

My research has particularly focused on the role of the upper airways during these techniques, with the goal of enhancing respiratory care for patients, and I consider myself to be very suitable support for Lisa's project. I am currently one of the world's leading physiotherapists in the field of chronic respiratory failure, including serving as the physiotherapy representative in a small expert group that organizes the multidisciplinary World Congress on Home Mechanical Ventilation (JIVD Congress), as well as the "Respiratory Failure and Mechanical Ventilation" course under the auspices of the world's largest respiratory organization, the European Respiratory Society (ERS). Through many years of work in an international environment, I have also developed a significant interdisciplinary network across several countries, where we actively collaborate on everything from clinical reasoning to drafting international guidelines.

After my PhD defence in 2018, I took on my first roles as a co-supervisor for two PhD students who successfully defended their theses in 2022 and 2024. Currently, I lead my own research group with four active PhD students, for whom I serve as the primary supervisor. This includes three physiotherapists and one medical doctor specializing in pulmonology. Additionally, I have supervised 10 master's students who have successfully completed their degrees and am

	currently supervising two additional master's students. Further, I have engaged as co-supervisor for Lisa Edel in her project in the planning phase.

Full Name	Dr Ema Swingwood
Organisation	UNIVERSITY HOSPITALS BRISTOL AND WESTON NHS FOUNDATION TRUST
Email Address	Ema.Swingwood@uwe.ac.uk
Role	Practice Supervisor
Description of Support Provided	Alongside my clinical consultant role, I am a Clinical Research Fellow at the University of the West of England (UWE) and a visiting lecturer at UWE and University College London for undergraduate and post-graduate Physiotherapy programmes. I have 20 years of clinical experience specifically in respiratory (both adults and paediatrics) and have been extensively involved in cough augmentation and airway clearance using the mechanical insufflation-exsufflation device, so have an excellent understanding of the clinical area that Lisa wishes to explore. I have lectured at International conferences over a number of years around these topics. Having completed my MSc in 2012, I have continued my research with my PhD work, which explored the use of MI-E in the intubated population. I currently hold an NIHR SCPRA award 2025-2027. I am the research lead for Diagnostics and Therapies at UHBW where a key component of my role is embedding research into the clinical setting and developing a research infrastructure for non-medics. My experiences with this will enable me to work effectively with Lisa during and beyond her PhD to develop a clinical academic role. I have supervised 9 MSc students (University College London) to completion, am supervising numerous ICA Internship applications, and am named as a clinical academic supervisor in another doctoral application. I have known Lisa for approximately five years, initially in a clinical capacity and through links within key national respiratory stakeholder groups, and more recently through informal mentoring. Lisa is a national expert in her field of the respiratory management of neuromuscular paediatrics patients. She has sought out opportunities to initially develop her research capability and research network. She is dedicated to optimising the care of her patients and her own development and I have no doubt that she will excel in this fellowship. I am thrilled to be able to support her and look forward to working with Lisa on this fascinating programme.

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Full Name	Professor Giovanni Baranello
Organisation	University College London
Email Address	g.baranello@ucl.ac.uk
Role	Primary Academic Supervisor
Description of Support Provided	I have had the opportunity to work closely with Lisa over the past 6 years as neuromuscular consultant at the Great Ormond Street Hospital and we have collaborated on several projects, like the correlation between a clinical respiratory score and a more objective respiratory assessment using the optoelectronic plethysmography. Lisa has a strong interest in developing research in the respiratory management of children with neuromuscular diseases and has brilliant ideas on how to demonstrate that the respiratory physiotherapy management of airway clearance can lead to improved respiratory status of children with Spinal Muscular Atrophy (SMA). Lisa has shown herself to be intellectually excellent, hardworking, well-organized, and has a keen and enquiring mind. She has excellent written and oral presentation skills and interacts absolutely well with colleagues. She has authored a few publications, she is actively involved in national and international research networks, like the International SMA Consortium, that include research centres from the UK, Italy and the US, and has presented her research projects at national and international conferences. I am committed to provide Lisa with all the necessary supervision during her PhD project, should she be successful in her application. I have the time and expertise to do so. As clinician academic I regularly provide supervision to postgraduate students. In the last 5 years I have supervised 5 PhD students, 2 as primary, and 2 have achieved completion. The mission of the UCL Great Ormond Street Institute of Child Health where I am based is to improve the health and well-being of children, and the adults they will become, through world-class research, education and public engagement. The UCL GOS ICH, together with its clinical partner Great Ormond Street Hospital for Children, forms the largest concentration of children's health research outside North America. The Dubowitz Neuromuscular Centre is the largest paediatric neuromuscular centre in the UK, and has a strong research interest in the causes and treatment of muscular dystrophies and other neuromuscular disorders in children. Along with myself, Professor Francesco Muntoni is also an academic member and lead of The Dubowitz Neuromuscular Centre. There are also other research members of the team that can provide support for Lisa's research project, including 3 research physiotherapists and 3 clinical research fellows. I recommend Lisa Edel unreservedly for this PhD application. She is outstanding in terms of her personal and intellectual attributes. I am sure that she will be an excellent PhD student and will progress to a successful clinical-academic career.

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Full Name	Dr Colin Butler
Organisation	GREAT ORMOND STREET HOSPITAL FOR CHILDREN NHS FOUNDATION TRUST
Email Address	colin.butler@gosh.nhs.uk
Role	Academic Supervisor
Description of Support Provided	I am delighted to write in support Lisa Edel and her application. The research proposal has the potential to significantly advance the care of children with Spinal Muscular Atrophy (SMA) where the focus of the research will be on understanding upper airway dynamics and how novel techniques can be applied to improve mechanical insufflation exsufflation in these children to improved respiratory health. I am an academic ENT surgeon at Great Ormond Street Hospital (GOSH)and a senior clinical lecturer and PI of the EpiCENTR group at the institute of child health at UCL. I have known Lisa for 2 years, both clinically and in a research perspective. At GOSH she is a specialist intreating children with SMA and her commitment to an academic focus will mean she will certain be a future leader in her field. The research proposal has the potential for high impact in improving respiratory health not only in children with SMA but has application to other pathologies. Lisa is in unique position at Great Ormond Street to investigate children with this severe airway disease and has an excellent track record for research to date. In terms of this proposal, I am extremely happy to offer assistance in supervising the clinical application of the endoscopy techniques of the proposed work and will assist Lisa with the analysis and research methodologies proposed. Lisa will have access to the Ear Nose and Throat (ENT) research group at ICH, where there are weekly research group meetings with airway-based research clinicians. ICH will provide Lisa appropriate postdoctoral research support, with appropriate training in place. My research group at ICH focuses on paediatric ENT research (clinical and laboratory based), where we have an active group of PhD and MSc students. I currently co-supervise 2 PhD students and 2 MSc students. I have completed supervision on 1 PhD student (completed within 4 years) and 2 MD students where they were awarded their degrees respectively. I also currently co-supervise a postdoctoral NIHR clinical lecturer in paediatric ENT. If awarded, Lisa will have regular supervision and be able gain peer support from the group of doctoral students linked to our group. A benefit of the cross-discipline approach will mean she will have support from several allied specialties. As an ENT consultant that focuses on airway research, I can provide clinical in training Lisa in endoscopy techniques required for the proposal and provide support for quantitative analysis proposed in this application, where our group has had experience in developing validated outcome measures to assess airway pathologies. In short I am enthusiastic in offering support to Lisa and this project that I believe to be capable of improving healthcare, not only for the targeted area, but in general for children with significant respiratory disease.

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Full Name	Ms Deborah Ridout
Organisation	UCL Great Ormond Street Institute of Child Health
Email Address	d.ridout@ucl.ac.uk
Role	Academic Supervisor
Description of Support Provided	As Director of the statistical support service at the Institute of Child Health, I have more than 20 years of experience collaborating with colleagues throughout the Institute, UCL and with external academic and industry partners. This has allowed me the opportunity to supervise and oversee many PhD and MSc students. In the last 6 years I have supervised 2 PhD students to completion and have been part of the supervisory panel for 6 others. One of my main areas of research is paediatric neuromuscular disease and I have collaborated extensively with experts in the field and have co-authored over 30 related publications. I have expertise in respiratory outcome measures in the patient population in the project and have supported many colleagues and students in understanding this type of data from a statistical perspective. I have helped design and set up many successful studies and am experienced in supporting and guiding students to develop their research skills. I am a statistics expert and can guide Lisa in her training and support her in applying her newly developed skills to her project. I have known Lisa for more than 6 years and found her to be a dedicated researcher who is keen and ambitious to explore new areas of research. I am committed to providing Lisa with all the necessary supervision during her PhD project, should she be successful in her application and have no doubt she will be an excellent student.

SUMMARY BUDGET

Justification of Costs

Travel and Subsistence

The budget focuses on two conferences, in alignment with the finance guidance. The costings relate to World Muscle Society conference and CureSMA. It allows for flights, £150 per night accommodation and £25 a day for subsistence for both conferences. Other planned attendance at conferences will be supported by other funding sources, for example the learning academy at Great Ormond Street Hospital. Within travel, this also accounts for 2 international visits to Bergen and Milan for training. It allows for flights, £150 per night accommodation and £25 a day for subsistence. It also allows for a visit to the National hospital for Neurology and neurosurgery.

Equipment

The equipment costs are in year one. The cost is for rental of equipment associated with transnasal flexible laryngoscopes. This is required equipment to complete the transnasal fibreoptic laryngoscopes described in work package 1. Leasing cost involves part purchase and part loan totally £28646 for a 4-month period VS. a purchase price would be £54350.40

Other equipment consists of an encrypted hard drive and memory stick, which is required for storage and back up of all project data. Cost is based on 2TB (£390) external hard drive and a 128GB memory stick (£70). A laptop has also been costed in the equipment section.

Consumables

The costs include:

- Cleaning of the Transnasal fibreoptic Laryngoscopes (£38 per scope for 50 patients = £1900)

- Double sided stickers for opto-electronic plethysmography: Single use stickers for the OEP, 52 stickers required per patients per visit = 2,600 stickers required (£60 per 500)
- Printing costs for posters for conference
- Printing costs for study related materials

Knowledge mobilisation, dissemination and impact

Costings include production of podcasts and talking heads production, to ensure equitable dissemination of information.

Training and development

Courses included in the career development plan are a combination of free and paid for courses. PPIE workshops and the statistics course are free to GOSH/UCL staff.

Paid for courses include;

- The Kings fund leadership course: The programme will equip leaders with the skills needed to manage a team, build resilience and respond to an ever-changing external environment.
- Bronchoscopy course at The Royal Brompton Hospital: A 2 day course for learning use of bronchoscopies (School of Bronchoscopy | Royal Brompton & Harefield hospitals)
- Beyond BASIC Airway Management Course at The Royal Brompton Hospital: To increase skills with management of the upper airway (Beyond BASIC Airway Management Course | Royal Brompton & Harefield hospitals)
- Impact Acceleration Unit, Knowledge mobilisation at Keele University: This is a fantastic opportunity for clinicians, academics and members of the public to improve their knowledge and understanding of the practical ways to design and use knowledge mobilisation theories and strategies in health and care practice.

Research inclusion

Courses related to research inclusion are included in the PhD programme and are free of charge. Access to talking mats to allow for all children to participate in consent and feedback. The annual fee is £120 for the complete package (https://www.talkingmats.com/new-digital-app-subscriptions/)

Working with people and communities

The costs relate to time, travel and subsistence for the committees involved in the project. Training is provided free of charge - Public Involvement in Research | Coursera. Include are also costings for refreshments for the study participants and families.

Other direct costs

The cost associated with this section is related to software required for upload of the videos from the new assessments. It is through ChilliPharm, a secure online platform. The cost includes 1 instance of access to ChilliPharm , 1 iPad filming kit to be used for the external patient videos provided for free, support with synchronisation of videos from the room and the TFL to allow for data analysis (per video synchronisation £400) with set up costs of £2500. Free access to online platform with ongoing support

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ACKNOWLEDGEMENT AND CONFLICTS

I have read and understood the terms of applying as Lead Applicant and accept this role	✓

Potential Conflicts	No conflicts to be declared

Narrative CV template for Personal Career Development Awards

****Lead Applicant / Co-applicant Name:****

Mrs Lisa Edel

****Module 1 – Contributions to the generation of new ideas, tools, methodologies or knowledge****

My interest in research developed into a passion, especially being part of a research active physiotherapy department and the neuromuscular service linked to the UCL Institute of Child Health, forming the Dubowitz Neuromuscular Centre of Excellence. I was fortunate to obtain an NIHR funded six-month internship, which provided the opportunity to further develop my research ideas. I have been interested in respiratory outcome measures to ensure meaningful patient benefit and clinical relevance. In 2016/7, I developed the Great Ormond Street Respiratory Score for SMA1 outcome measure which monitors ventilation and dependence on respiratory physiotherapy. I presented this at the World Muscle Society Conference in 2018 and was awarded the Lea Rose Award for best SMA presentation. This work is published (DOI: 10.1016/j.nmd.2021.01.008) and since implementation has enabled better consistency of care with measurable respiratory function. It has been utilised in Brazil and Poland and cited in nine other publications. I presented the longitudinal data at the World Muscle Society Conference in 2023. My skills in clinical research continue to grow, writing two manuscripts as first author. The first is a scoping review on physiological effects of mechanical insufflation exsufflation in patients with neuromuscular disease and the second is a paper developing a global outcome measure for respiratory function in SMA, which are both in the process of publication in respiratory care and journal of neuromuscular diseases respectively. This has given me key skills of critically appraising the literature, reviewing its quality using quality appraisal tools, as well as improving my writing skills. I continue to develop the research aspect of respiratory management of neuromuscular patients, within a focused research position. This role has two main aims: improving respiratory care for patients with neuromuscular conditions and developing and improving national and international respiratory neuromuscular networks. I am currently the principal investigator on a research trial looking at comparing optoelectronic plethysmography to standardised lung function methods.

****Module 2 – Contributions to the development of others and maintenance of effective working relationships****

Education is a key component of my role and as I develop to become a leader in my field. I provide training to other professionals and students through my clinical role as well as on MSc modules at University College London and externally through the European respiratory society and pharmaceutical partners. I am committed to developing and skilling up the next generation of physiotherapists and want to ensure high quality knowledge translation and supervision. I have recently supervised two BSc students, two peers as well as supported members of my profession nationally and internationally. I have supported them through building their research ideas, highlighting the importance of data collection, helped with conducting audits, service evaluations and quality improvement projects. This aims to increase clinical and research capacity and capabilities. The support has provided patient benefit by translation into improved management and outcomes of care, including a nationwide review of mechanical

insufflation exsufflation use on paediatric intensive care and a review of airway clearance techniques used by patients with Duchenne's muscular dystrophy. I hope to be a research champion for my profession and encourage other allied healthcare professions to engage with research. I have jointly led an international collaboration with colleagues, setting up a panel of respiratory specialists to gain consensus on effective respiratory care for SMA. This has involved colleagues in the UK and I have built collaborations with colleagues in Europe and the USA. This has improved our understanding of international treatments, standardising care internationally. As a highly motivated clinical specialist respiratory neuromuscular physiotherapist, my career has enabled me to become a recognised expert in my field, developed highly specialised clinical skills, lead on service developments, deliver taught master classes internationally, participated in and created my own research and remain driven to achieve in a clinical academic career.

****Module 3 – Contributions to the wider research community****

In my current clinical role, I am fortunate to have one and a half days a week to focus on research. This was developed through a successful application to the pharmaceutical company, Roche. As part of this role, I am able to lead on developing a strong research work ethic within the allied healthcare professionals at Great Ormond Street Hospital, as well as develop ongoing research activities with collaborators in Europe. As part of my ongoing commitment to research development and the study I am currently running, I have created animated videos to widen accessibility to aid the consent process. These animated videos were designed to promote accessibility by ensuring clear understanding of the research for all patients, helping to allow better inclusion of patients from diverse backgrounds. The videos were accompanied by a range of patient information leaflets for a variety of levels of understanding. The combination of these approaches was designed to allow for equality for all participants. In addition, I have established a national respiratory neuromuscular network, which has informed guidelines, ensuring consistency in practice, improving patient care. I have initiated and led a workshop for all professionals involved in the respiratory care of children with SMA. This has resulted in an increased network for patient care and collaborative research. Alongside this, I have contributed to national evidence-based guidelines for Duchenne's and SMA. I continue to contribute to community events, being an invited speaker at international conferences, such as JIVD 2025, and am an invited speaker at an SMA UK led conference in Nottingham in September2025.

****Module 4 – Contributions to broader society****

As the junior faculty representative for the NIHR-BRC Patient and Public Involvement and experience(PPIE) steering group, I have been part of guiding how we offer support, train, and develop HCPs within the research cycle. We have also looked at methods, and development of toolkits and templates to help capture the impact of patient and public involvement. As part of this role, I have supported the Children and Young Persons forum as well as led a webinar on 'What makes a good plain English summary'. I work alongside Muscular Dystrophy UK and Spinal Muscular Atrophy UK charities to provide education, extending my reach to special interest groups, national and international meetings. I am invited to speak internationally on respiratory care for spinal muscular atrophy (SMA) to multi-disciplinary teams, further increasing opportunities for education and collaboration. I have good working relationships with pharmaceutical companies as well as other private companies, including online

digital platforms, e.g. ChilliPharm and medical device suppliers, e.g. Breas, BTS bioengineering. I have had the opportunity to be involved with their international teachings, reaching other clinicians, advocacy groups and families. This enables improved practice across the combined sectors.

****Module 5 - Your suitability for this award****

This award is the essential next step to build upon the research skills I have developed to date, acquire new skills and knowledge, providing the foundation to embark on a clinical academic career. The goal is to develop a programme of clinical research, leading on research informed respiratory outcomes, to enhance patient quality of life, advocate for improved care and influence policy and public engagement.
This award will provide me with the opportunity to develop a comprehensive understanding of research designs, methodologies, and enhanced statistical analysis skills. It will facilitate focused time on equality, diversity, and inclusion; working with people and communities, ensuring research protocols reflect patient focused outcomes sensitive to the diversity of populations. The enhanced leadership skills developed during the award will facilitate skilled management of the complex patient population as well as support the development of future clinical and academic physiotherapists in this highly specialist area.
The training will enable me to work with the supervisory team, mentors, and wider neuromuscular community to develop as a clinical academic, as the focused research project is undertaken. The links with international supervisors will enable me to continue to develop and improve national and international respiratory neuromuscular networks as well as form links with charity and advocacy groups.
I am passionate about achieving health improvements for the children I have looked after for the last 10 years and by completing this research, the goal is to answer previously unanswered and under-researched questions. The planned research will highlight key areas for development within this patient group as well as show broader potential for other patients with respiratory compromise, which I plan to take forward in future research projects such as a post-doc.
My career development as a clinical academic will be supported by a strong research culture within Great Ormond Street Hospital, of research active Allied Health Care Professionals. This infrastructure includes the BRC which funded my internship and provides ongoing support to early career researchers, from a wide range of clinical backgrounds. I have already benefited from the particularly strong cohort of clinical academic physiotherapists, collaborating on a scoping review, joining and partaking in teaching sessions with the clinical academic team as well as ongoing supervisions. They will support my journey to become a clinical academic and have enabled consultancy level posts within research. I will also benefit from links with the Centre for Outcomes and Experience Research in Children’s Health, Illness and Disability at Great Ormond Street Hospital. Working at a research active hospital, where every bed is considered a research bed, embeds a proactive research focused culture. It is the ideal place to embark on a clinical academic career. I plan for this award to be the beginning of a long career as a clinical researcher.

Detailed Research Plan

Background and rationale

This research focuses on improving respiratory management in children (0-18years) with Spinal Muscular Atrophy (SMA), aiming to revolutionize management.
SMA is an autosomal recessive neuromuscular disorder caused by the deletion of survival of motor neuron protein 1 (*SMN1*), which is responsible for 90% of SMN production in healthy individuals (1). Another gene, survival motor neuron 2 (*SMN2*) produces insufficient protein to maintain muscle function. Due to the lack of SMN protein, SMA is characterised by degeneration of spinal cord anterior horn cells leading to muscle atrophy, proximal muscle weakness, respiratory and bulbar impairment (2).
SMA affects all ethnicities, with higher mutation rates in Caucasian and Asian populations (excluding Asian Indian), and in the United Kingdom has an incidence rate of 1 in 11,000 live births and a prevalence in the UK of 1-2:100,000 (3). SMA is classified into 3 main subtypes (Type 1-3) based on first clinical manifestation and SMN2 copies, with the fewer copies correlating to the most severe (1). SMA Type 1 is the most common form, presenting in the first six months of life with progressive muscle weakness and bulbar (swallow) dysfunction with resultant chest infections and respiratory failure, leading to death before the age of two (2). SMA Type 2 are able to sit but never walk and ultimately require respiratory support (4), whilst SMA Type 3 are able to walk initially, loosing function over time. They may require respiratory input at a later stage of the disease (4). Across the SMA spectrum, respiratory failure is the major cause of morbidity and mortality(5). Standards of care, supportive therapies and the introduction of disease modifying treatments (DMT) have driven an interventional approach and have improved life expectancy (5). SMA has been added to many newborn screening programs worldwide and has changed the disease landscape (6). However, due to a lack of newborn screening in the United Kingdom, 47% of patients already demonstrate clinical signs of the disease when receiving DMT, specifically respiratory compromise (6). SMA has shifted to a chronic life-limiting condition (7) with children living with the disease, attending school, and participating in the community. Regardless of clinical presentation, respiratory impairment is the key clinical sign to tackle across the wide spectrum of SMA.
Children with all types of SMA have difficulties with secretion clearance due to weak respiratory muscles and an ineffective cough, resulting in mucus plugging, impaired mucociliary clearance and lung collapse (8). Effective secretion management and respiratory physiotherapy play a crucial role in preventing respiratory infections, reducing hospitalisations, and improving life expectancy and quality of life (2). These patients require reviews with a specialist physiotherapist and individualised treatment programmes, including one specific treatment; mechanical insufflation-exsufflation(MI-E), commonly called a cough-assist machine (8). This device mimics a cough by delivering a positive pressure breath to encourage deep inhalation (insufflation), followed by a rapid switch to a negative pressure breath (exsufflation) creating a vacuum and subsequently increasing expiratory flow rate. This moves secretions higher in the oropharynx allowing effective clearance. MI-E has been recommended by international standards of care as a vital treatment for patients with SMA to improve patient outcomes (4). Despite DMT targeting motor and respiratory function for SMA, optimizing respiratory management with MI-E remains essential to patient care. A significant limitation is there is no consensus for optimal MI-E

pressure settings, with clinicians rely on observation and patient feedback to establish appropriate pressures, lacking clear evidence-based guidance (9). Patients with SMA and bulbar dysfunction may face more challenges in utilising MI-E due to degeneration in corticobulbar and brainstem pathways (7). Research in amyotrophic lateral sclerosis (ALS), shows therapeutic pressures with MI-E may cause a disadvantageous laryngeal response, reducing cough efficacy (10). Furthermore, the recent completed scoping review (Edel et al, article in press) concluded there is limited consistent evidence into the effects of MI-E on airway dynamics and lung function. Further investigation is required exploring the specific MI-E settings and the impact on physiological effects, including the upper airway response. Studies specifically focusing on a paediatric neuromuscular cohort are vital, as this is an under researched area.

This research aims to discover the airway response to MI-E, including the potential adverse reaction with appropriate optimisation, highlighting the physiological effects to MI-E in terms of airway dynamics and lung function. This in term would release the potential for improve care models, directly benefit the secretion management of the patient, reducing infections and hospitalisations.

The exploration of this topic has been identified by the James Lind Alliance, under the SMA Top 10 priorities, number 5 - What is the best physiotherapy approach for people with SMA? Does it need to be tailored to the individual (e.g. according to age and type of SMA)? Does it need to change during treatment with one of the new drugs?

Improving respiratory outcomes: This research aims to determine if the airway status of patients with SMA, measured using Transnasal Fibreoptic Laryngoscope (TFL), can affect the delivery of effective MI-E pressures. The effectiveness of MI-E pressures can be measured by differences in lung expansion, reviewed by utilising opto-electronic plethysmography(OEP).This may lead to reduced risk of lung collapse, improved respiratory function, decreased chest infections, fewer hospital admissions, directly impacting morbidity and mortality in patients with SMA (11).

Reducing hospital admissions: Despite international implementation of DMT, global hospitalisations remain high due to respiratory complications in SMA patients. Approximately 20% of life of a child with SMA is spent in hospital, with an estimated lifetime cost of £2.2M (12). Similarly, in the United States, each admission for a child with SMA1 costs appropriately $50,190 and they require an average of 4.2 hospitalisations annually (11). Optimal respiratory management is critical as respiratory complications remain the primary cause of mortality and morbidity, despite disease modifying treatments with is a large physiological, psychological, educational, and economic impact to the patient and family (13)

Supporting development of care pathways: Current standards of care (4) highlight importance of MI-E for SMA patients (2, 8). Limited data is available to guide optimal settings and frequency of use (8). This research will guide device settings aiming to create clinical guidelines, identifying the safest and most effective parameters for MI-E, which would be of benefit to clinicians, patients, and future clinical trials.

Use of TFL: The larynx controls the opening and closing of the airway, with precise co-ordination. Utilising TFL is the gold standard test for visualising the upper airway, enabling clear documentation of the sequencing of the airway dynamics. It is well tolerated and used successfully in adults (10). Airway clearance with MI-E for patients with SMA is highly complex and has life changing results when used effectively (8). Literature in adults with ALS suggests MI-E treatment failure is likely due to incorrect sequencing of the airway dynamics during insufflation (14). This

response prevents air filling the lungs during insufflation, causing discomfort and subsequent inefficient exsufflation. This means MI-E is not effective for all patients. Existing literature reviewing upper airway dynamics in adult ALS (14) do not enable conclusions to be made regarding clinical translation to SMA, highlighting the importance of the proposed study.

The community representatives highlighted some concerns regarding insertion of the TFL. However, when the size of the camera was highlighted as being smaller than suction catheters they use day-to-day, it was viewed as acceptable. The patients and parents suggested including patient videos and step-by-step guides to help children understand the test. These have subsequently been created and are available for patients, parents and carers to view(GOSH_GLA_BreathingAnimation - YouTube).

Use of OEP: SMA children have breathing asynchrony and more restrictive chest walls, leading to respiratory complication. This is particularly seen in patient with SMA1 (15). To guide care, lung function tests are usually performed from six years of age, requiring active participation. In SMA children, these tests are challenging due to age and muscle weakness (15). OEP investigates chest wall movements and volumes, by measuring quiet breathing, inspiratory capacity and voluntary cough, without the need for patient cooperation. It represents a safe and well tolerated solution for patients (15). Encouraging preliminary data in neuromuscular diseases has been published, highlighting the need for further investigation (22).

Patient comfort and tolerance (MI-E and new tests): Patient tolerance and comfort greatly affects user compliance (9). One study focusing on children's tolerance to MI-E pressures, found personalised settings to have better comfort scores when compared to symmetrical or asymmetrical settings (9). It will be key in the current proposal to gain feedback on levels of comfort, tolerance ad acceptability of the new tests as well as if the patients feel increased comfort with optimised MI-E settings.

Research Question: How do airway dynamics change in children with Spinal Muscular Atrophy during assisted coughing?

Aims and objectives

Aims:

1. To investigate reasons for variability in effectiveness of MI-E in children, establishing optimal settings for MI-E in patients with SMA
2. To assess variations in lung function with optimised MI-E settings
3. To evaluate tolerance and acceptance of new assessment techniques.

Objectives:

Objective	Outcome
1. Utilising TFL to assess the upper airway in children with SMA	Direct visualisation with TFL of the upper airway dynamics
2. Evaluating to optimise the upper airway response patterns to MI-E, using TFL.	Comparison of upper airway dynamics on varying MI-E settings with direct visualisation
3. Assessing changes in respiratory patterns/lung volumes utilising OEP with baseline and optimised MI-E settings.	Comparison of lung volumes on baseline and optimised MI-E settings with OEP
4. Measuring patients/carers tolerance and acceptability of new assessment techniques through competition of a visual analogue scale, with adaptations made for age and cognitive ability.	VAS throughout use of the new tests.

Methodology / Plan

This is an interventional quantitative study to investigate upper airway response to MI-E in children with SMA. Following the planning phase (months 0-9), this project will be divided into 3 main work packages (WP) completed over a 36-month study. Time frames are included in the GANTT chart. WP1 will address objectives 1 and 2, WP2 will address objective 3 and WP3 will address objective 4 as described above.

Planning Phase

The initial nine months of the award encompasses career development training, alongside gaining ethical approval and local research/development authorisations.

Working with patients and communities in the planning phase

Building on work completed by the applicant in the planning stages of this proposal, the families and community representatives will be asked to join online meetings to discuss and feedback on preliminary questions regarding comfort, tolerance, and acceptance of the new proposed tests. They will be asked how questions could be modified to the most suitable way of asking children and families.

During earlier work with the community and in discussions with groups who have helped shaped this proposal, online meetings were requested. They also asked that the families were separated into functional level of SMA. This is related to difference responses to treatments and the sensitivity of those families.

Following these meetings, a list of questions will be confirmed, which will be asked to patients during the new assessment techniques. These will be scorable (for example on a visual analogue scale) and adaptable for all patients.

Power Calculation for all Work Packages

The use of the OEP in a cohort of 7 SMA patients was tested to assess usability in December 2023. Patients were requested to participate in testing of the set-up of the OEP laboratory alongside their existing appointments. The patients were requested to lie flat, breath at tidal volume and use their MI-E while being tested. Each patient was assessed in a 30-minute period, and the testing was well tolerated.

Unpublished data collected during the initial laboratory set-up allowed initial analysis to demonstrate a clinically significant meaningful change in respiratory insufflation volume of 1.5mls/kg when adjusting MI-E pressures. With a standard deviation of 3.2, we would need 36 patients to account for this difference as statistically significant with 80% power and 5% significance. Allowing for 10% attrition (patient drop out, illness and death), we aim to recruit a total of 45 patients from all functional SMA groups to ensure we recruit at least 36 patients. Our current pool of patients aligned with the inclusion criteria is 85 patients.

<u>WP1: TFL for airway assessment and during MI-E with children with SMA</u>

This aims to answer objectives 1 and 2.

This WP is divided into 2 stages, a) baseline assessment and b) evaluating upper airways with alterations to MI-E.

a) *TFL baseline assessment*

<u>Identification and Recruitment:</u> Potential participants will be identified through clinics at Great Ormond Street Hospital(GOSH). To limit potential bias, patients will be informed of the study and given a patient information leaflet from clinic staff, during routine hospital visits, who are not involved in the study. Participants will be invited by telephone and a follow up email to take part. We will pursue random sampling. They will have opportunities to ask questions and watch a specifically developed

animation explaining the study. All participants/caregivers will be asked to provide informed assent/consent.
Inclusion criteria: Genetic confirmation of SMA, use of MI-E for airway clearance for >3 months, care under respiratory physiotherapists at Great Ormond Street Hospital, including children of all age groups (0-18years).
Exclusion criteria: Patients on palliative care pathways and/or patients who have transitioned to adult services.
Methodology: A cross-sectional study.
Clinical baseline data will be collected on a specifically designed template will include (utilising existing databases):
Age; Date of birth; Gender; Ethnicity, Growth centile (Weight/Height); Copies SMN; SMA Treatment and age at treatment; Non-invasive ventilation (NIV) user/hours of ventilation; Chest physiotherapy techniques/nebulisers; Medications; Admissions and antibiotic use in the last six months; Compliance data to MI-E /compliance to NIV; Existing secretion management plan/baseline MI-E pressures and start date; Functional ability/motor function score; Scoliosis (Cobb angle) +/- surgery.
Procedure (estimated length of procedure 30 minutes)

- On arrival, patients will be able to ask questions regarding the tests/equipment.
- They will then be placed in a semi-reclined position.
- Patients will be monitored for oxygen saturations throughout the assessment.
- The laryngoscope will pass through a modified interface which attaches to the MI-E, and then placed on the patients face.
- The laryngoscope, which is 2mm in diameter, will be introduced to the patient trans-nasally.
- The examination will be visualised on a screen with continuously running video recordings from the TFL output and an external video camera recording the screen of the MI-E device, allowing for documentation of phases of insufflation (breath-in) and exsufflation(cough).
- Video-recorded TFL will visualise laryngeal anatomy at rest (normal breathing) and during baseline individualised patient MI-E settings (as determined prior to start of the study)
- Different levels of the upper airway (Glottic, Supraglottic and Hypopharyngeal) will be scored (0 abduction/open, 1 adduction/closed) documenting the outcome of laryngeal responses to MI-E.
- During the exsufflation, cough peak flow will be recorded through a Pneumotrac 2 spirometer.
- Using questions agreed through working in partnership with the community related to tolerance, comfort and acceptability, all patients will be asked to report levels of comfort and tolerance of the TFL test both during normal breathing and when using baseline MI-E settings, on an appropriately measurable score, for example, a visual analogue scale or an adapted version for children of all abilities.
- Other parts of equipment set up: a video camera monitoring of the patient and, a pneumotachograph and pressure transducer in the MI-E circuit to give ongoing flow- and pressure tracing.
- Inter-rater reliability will be measured through review of a random selection of recordings by a blinded assessor (within the supervisory team).

Timeline: Months 9-18 (Year 2 – refer to Gannt Chart)

Data Collection and Analysis: Data collection will be targeted over 9 months, with the goal of testing 8 patients per month, maximum 2 patients per day. Data collection takes approximately 1-2 hours per patient. Data will be collected on an existing evaluation form from previous work to allow for review of each level of the larynx and appropriate analyse (10). It allows for comments and documentation to indicate the stage of MI-E cycle.
Anonymised videos will be uploaded to ChilliPharm, a secure online platform specialising in clinical trials, and videos obtained from TFL will be uploaded alongside the video monitoring the pressure on the MI-E device. The videos will be synchronised by a verbal cue, screened, and analysed by the principal investigator. The online platform also allows for inter-rater reliability testing.
Data will be recorded in Excel, coded and transferred to R for statistical analysis. Baseline characteristics will be analysed using means and standard deviations, or number and proportion as appropriate. Between group demographic analysis based on functional levels will be analysed using ANOVA, Kruskal Wallis tests or Chi Square test.
Airway status will be analysed as abducted or adducted at each level of the upper airway and compared by the recognised SMA functional groups (non-sitters, sitters, standers, and walkers combined) and analysed utilising Chi squared or Fishers exact test. The lowest open level with be compared between functional groups using Mann Whitney tests.
Incomplete data will be excluded.

b) ***Evaluating upper airways response patterns with MI-E and whether changing MI-E settings alters response patterns.***

Identification, recruitment, inclusion, and exclusion criteria are as identified in part a) of WP1. The only additional exclusion would be non-tolerance to TFL from part a).
Methodology (estimated length of procedure:30 minutes)

- An interventional pre and post study, examining effect of changing MI-E settings on the upper airway responses.
- Patients will start on pressures identified in the baseline phase.
- Following a protocol-driven stepwise approach, created for the purpose of this study based on previous work (10), settings will be adjusted to obtain highest pressures with an open upper airway.
- If the airways are open at baseline, MI-E device settings will be increased in 2cmH2Oincrements until airway closure is noted.
- The settings will then be reduced to the last open level documented.
- If the airways are closed at baseline, MI-E settings will be decreased in 2cmH2O increments until airway opening is seen.
- These settings will be documented as the patients' optimal settings.
- This provides data to characterise the upper airway at different levels of positive and negative pressure.
- TFL assessment together with flow- and pressure tracings, analysis and measurement of inter-rater reliability are performed as in the baseline phase.
- Using questions agreed through working in partnership with the community related to tolerance, comfort and acceptability, all patients will be asked to report levels of comfort and tolerance of the MI-E settings during each change, on an appropriately measurable score, for example, a visual analogue scale or an adapted version for children of all abilities.
- The key questions will focus on optimal settings and when airway closure is seen.

Timeline: Months 9-18 (Year 2 – refer to Gannt Chart)
Data Collection and Analysis: Data collection will occur simultaneously with part a) of WP1 and will occur at a range of settings on the MI-E device, during one session unless patients/caregiver request a rest periods.
Airway status will be analysed at each level of the upper airway at the different settings with clear documentation of the maximal settings. The maximal setting between groups will be analysed utilising a T-test or Mann-Whitney test.
Cough peak flow from baseline settings and maximal modified settings will be compared and analysed with paired t-test or Wilcoxon matched pairs test.
Data will be analysed for relationships between age and SMA functional levels.
Success Criteria WP1 A and B:
Objectives:

- To determine the laryngeal response patterns to MI-E in a paediatric SMA cohort.
- To gain physiological understanding of airway status of SMA children at rest and on utilising MI-E.
- To determine maximal airway pressure from MI-E with all levels of airway open compared between functional groups.

Risks:
Recruitment/retention of participants.
Anxiety related to unknown tests.
Issues concerning equipment availability and malfunctions
Mitigation of risks:
Timely consent with adequate resources i.e. videos to explain processes
Working with charity partners to raise awareness of the study.
Equipment available at time of consent
Appropriate time for assessments
Equipment will be purchased for the study with backup devices
Previous use in a small paediatric cohort in Norway with no major complications and it was well tolerated.

WP2: Assess changes in respiratory patterns/lung volumes utilising opto-electronic plethysmography(OEP)

This aims to answer objective 3.
Identification, recruitment, inclusion, and exclusion criteria are as identified in WP1. The only additional exclusion would be incomplete data from WP1.
This is a follow-on plan from WP1, completed on the same day after a break, as suggested by working with patients and the community.
Methodology (estimated length of procedure: 30 minutes)

- In the existing OEP laboratory at GOSH, the child will be positioned in supine on a plinth with their thorax and abdomen free of clothing.
- Temporary reflective markers for the OEP will be placed in specific locations, according to anatomical landmarks and existing protocols, throughout the thorax and abdomen.
- They will be distracted/watch television to remain relaxed.
- The temporary markers are picked up by infrared cameras with signals translated through computer software to recreate the shape and movement of the chest wall.
- Data will be obtained whilst the child is completing:
    - Spontaneous quiet breathing for 5-minutes
    - A spontaneous cough
    - Baseline settings of MI-E

    - Quiet breathing for 5-minutes (washout phase between settings)
    - Newly adapted MI-E settings
    - Further 5-minute period of quiet breathing
- Using questions created through working in partnership with the community related to tolerance, comfort and acceptability, all patients will be asked to report levels of comfort and tolerance of the OEP on an appropriately measurable score, for example, a visual analogue scale or an adapted version for children of all abilities.
- The question on comfort will also focus on the difference between MI-E pressures.

Timeline: Months 9-18

Data Collection and Analysis: Data is recorded through the computer programme EVO. It will be analysed using a programme called SMART TRACKER. The software allows for the following data collection during all tests:
Tidal Volume(TV), Respiratory Rate(RR), Forced Vital Capacity(FVC),Forced Expiratory Volume(FEV1), Total lung capacity(TLC),compartmental chest wall volumes and Cough peak flow(CPF).
Data collected will show lung volumes on the original MI-E settings and the new settings which were amended under direct visualisation with the TFL. This will enable researchers to establish if personalised pressures allow for improved lung function, translating in to improved airway clearance and reduced hospital admissions.
Respiratory function data from original MI-Es settings will be compared to optimised settings and will be analysed using a paired t test or a Wilcoxon matched pairs test.

Success Criteria WP2

Objectives:
Assess changes in respiratory patterns/lung volumes utilising opto-electronic plethysmography(OEP) with baseline and optimised MI-E settings.

Risks:
Poor tolerance to lying flat.
Lack of understanding of the requirements during the tests
Fatigue from day of testing
Safeguarding for children to be free from clothing on their chest and abdomen

Mitigation of risks:
Equipment modification/additional supports allowing for enhanced patient comfort.
Suitable distraction techniques
Parental support during tests with adequate rests to reduce fatigue
Suitable safeguarding measures to be in place

WP3: Evaluation of tolerance and acceptance of new assessment techniques.

Identification, recruitment, inclusion, and exclusion criteria are as identified in WP1a.

Methodology

- Using the answers from the questions asked throughout WP1 and WP2 related to tolerance, comfort and acceptability, data will be extracted, allowing for SMA functional status.
- The scores relating to tolerance/comfort/acceptability of procedure (TFL and OEP) as well as scores relating to comfort level of MI-E on baseline and optimal settings, will be in either a number format or assigned a number for data analysis.
- Scores regarding tolerance, acceptance and comfort will be analysed to gain patient perception.

Timeline: Months 9-18 (Year 2 – refer to Gannt Chart)

Data Collection and Analysis: Data collected for the new assessments will be grouped for tolerance, comfort and acceptability and analysed using means and standard deviations for each group. The groups will then be further sub divided to allow for analysis in functional groups. Comfort scores related to baseline settings and optimised settings on MI-E will be analysed using a paired t test or a Wilcoxon matched pairs test.

Success Criteria WP3

Objectives: Measure patients/carers tolerance/acceptability/comfort of new assessment techniques.

Risks:

Lack of understanding or ability to report comfort/tolerance/acceptability

Mitigation of risks:

Adapted scales to allow for verbal/non-verbal communication

****Knowledge mobilisation, dissemination and impact****

The intended outcomes are linked to establishing how the upper airway responds to MI-E with patients with SMA. The key stakeholders/knowledge users are the patients, families, and carers, as well as healthcare professionals working with patients with SMA. It also has implications for the wider neuromuscular community as the results may influence other conditions.

Parents/Patients/Charity dissemination

Parents and patients highlighted importance of sharing feedback. They have provided feedback for study design and will continue involvement within the patient and community steering group. They provided insight into preferred dissemination methods. We will ensure allowances are made for those people whom English is not their first language. Knowledge mobilisation will occur throughout the project, ensuring stakeholders are kept up to date with project developments. Some methods are highlighted below:

- Lay summary for parents/carers and patients in a leaflet/infographic format via coffee mornings, noticeboards, and parent conferences
- Online reports for relevant charities and GOSH websites
- Podcasts/Talking heads presentations.
- Participation at GOSH/BRC research events
- Translation into a variety of languages (e.g. Polish/Arabic as common languages in this population)

SMA UK and MD UK will support dissemination by sharing findings on websites and social media, engaging families from a diverse range of backgrounds.

Conference presentations and publications

Conferences will be targeted to reach families and multi-disciplinary audiences and include: Cure SMA (2026), International Conference on Home Mechanical Ventilation and European Respiratory Care Association Congress (2028), SMA Europe (2028), World Muscle Society (2028).

A minimum of four international peer-reviewed journal articles are planned, ensuring multi-professional readership. Targeted journals include Thorax, American Journal of Respiratory and critical care medicine, and Neuromuscular disorders.

International influence

Existing collaborations with international charities, advocacy groups and networks for SMA with ensure international output of study results, improving dissemination. This ensures information and results will be appropriately fed back and are widely accessible to all target audiences, maximising impact.

Impact
MI-E is used in neuromuscular patients as well as in neurodisability (for example, cerebral palsy and hypoxic brain injuries) and in the intensive care setting. A constant challenge for clinicians is set up of MI-E devices and pressures required for optimal airway clearance. Utilising TFL will give us information regarding the upper airway of these patients and a range of parameters for MI-E settings. Personalised settings for MI-E would reduce risk of dynamic airway collapse during treatments. Clinically, the new assessments would be implemented into clinical practice, allowing future prescribing of MI-E to be more specific, improving secretion clearance, reducing risk of infections, and reducing dependence and associated costs on NHS services. In turn it will improve the health and wellbeing of the child. In addition, to reach more patients, a clinic would be set up for all patients to be referred to for complex airway clearance with MI-E. It reduces potential time spent in hospital and the associated disruption to patients, parents and family members, including education and work commitments.
This project may translate to other conditions requiring MI-E, helping patients as well as providing evidence on safety and efficacy of MI-E. It enables the professional community to agree consensus on MI-E setting prescriptions where there is a gap in understanding, impacting on future policies and guidelines.
For the first time in the United Kingdom, children with SMA will have an alternative to traditional spirometry and this project will allow for direct measures of lung volumes utilising OEP. This enables improved treatments and reduce time spent in hospital.
Future study designs
Future work will include 1) a patient reported outcome measure focused on patient comfort and tolerance for respiratory physiotherapy techniques and 2) a comparison study between SMA patients who follow standards of care compared with MI-E adjusted under direct visualisation with TFL, utilising OEP to highlight between groups results. This would be planned to be undertaken as part of an advanced fellowship or an application to the Research for Patient Benefit Programme.

****Project management****

Research will be conducted in line with the Declaration of Helsinki and Good Clinical Practice. The Gantt chart provides oversight of the project. Additionally, the university provides milestones for specific targets with set timeframes (promotion to higher award at end of year 1 and thesis committee meetings). This will provide oversight from academic perspective. Alongside this, monthly meetings with the supervisory team will ensure the project is completed in the planned timeframe. IN addition to the supervisory tea, a steering group will be in place to offer strategic and operational advice. I will be leading on ethical and local approval, risk assessment, data collection and analysis. I will be first author on all outputs and any conference presentations. Success criteria has been added to each work package.

****Intellectual property and commercialisation****

Intellectual property and commercialisation will likely not apply to this research project. No patents or copyrights will be created. Patient and public benefit will be supported by reviewing current practice and assessing potential adaptations to current treatments.

Career Development Programme for Personal Career Development Awards

****Training courses****

Academic Training: University College London (UCL) will host the PhD and has a world leading reputation and registration on the graduate research development programme will provide access to an extensive range of online resources and taught courses. My training and development plan has been created using the VITAE Researcher Development Framework, which ensures my research needs are met within all four domains. These encompass knowledge and intellectual abilities, personal effectiveness, research organisation and governance, and communication and impact. The formal and additional academic training, outlined below, has been developed to cover all aspects of work package one to three. This has been discussed with the supervisory team and the courses listed are felt to be the best courses offered for to support the methodological training and PhD journey.

Formal Academic Training - part of the UCL PhD programme

Year 1

- Networking and Collaboration (Domain D)
- An introduction to R with RStudio (Domain A)
- Excel advanced large data collection (Domain A)
- Introduction to critical appraisal of quantitative health research (Domain A)
- Facilitation: how to get the most out of your PPI activities (Domain D)
- The Writing Series: Academic Writing logic (Domain D)
- Introduction to Research Integrity and Support (Domain C)
- The PhD Journey (Domain B)
- Time Management (Domain B)
- Research Ethics (Domain C)

Year 2

- Engagement and Impact: evidencing and resourcing for impact (Domain D)
- Managing research data to unlock its full potential (Domain C)
- Writing your PhD thesis (Domain D)
- Focus on Peer Review writing and publication (Domain D)
- Getting more out of EndNote (Domain A)

Year 3

- Academic Career Planning for PhDs (Domain B)
- Podcasting made easy (Domain A)
- Networking Skills (Domain D)
- Researcher's skills beyond academia: leadership (Domain B)
- Finding Funding Opportunities (Domain C)

Additional Academic Training: Opportunities will be specific to the project. Additional Academic training in year 2 will be limited due to high demand for data collection.

Year 1

Good Clinical Practice - a refresher course would be taken to protect the rights and safety of the participants and collect reliable research data

Statistics courses - Chi-square and Beyond for 2x2 Tables at UCL – 1 day. This course is designed for healthcare professionals who already have some understanding of basic statistical principles and wish to develop their statistical

expertise. I will learn specific in depth analysis regarding chi-squared analysis which are statistical methods vital for this study and studies in the future.
PPIE training workshops, UCLH – each workshop is 2 and a half hours. This course has a variety of workshops which enables a broad understanding of PPIE. It will ensure I link all aspects of the project to PPIE, allowing for feedback and adaptations as required, supporting the development this and future projects. It will also ensure I get the best results from the PPIE undertaken, including that of the steering group and other PPIE activities.
Year 3
Leadership Course: Emerging Clinical Leaders, The Kings Fund, 6 days. This course is renowned for supporting clinical leaders. It will provide me with strategies to help lead future clinical academics.
Impact Accelerator Unit: Knowledge Mobilisation Course: Keele University, 3 days. This is renowned course to improve knowledge and understanding of the practical ways to design and use knowledge mobilisation theories and strategies in health and care practice, which will ensure effective implementation of research findings and impact for patients, now and in the future.

Clinical Training: Throughout the fellowship in addition to my academic training, I will develop advanced clinical knowledge and skills. Planned training is below:
Year 1
Beyond Basic Airway Course at Royal Brompton Hospital – 2 day course
Bronchoscopy course – 2 day course
These courses are renowned for teaching enhance airway skills which will be essential foundation blocks for WP1. These will then be further developed with Tiina Andersen on placement.
Clinical supervision with Colin Butler, Ear nose and Throat Surgeon at Great Ormond Street Hospital, providing joint sessions, teaching and clinical supervision.
Year 2
Reciprocal visit from Dr Andersen for supervision in own paediatric setting
Year 3
Maintenance of advanced clinical skills and competence with on-going clinical supervision with Colin Butler (Joint clinical sessions/case study reviews)

****Placements****

Year 1
Clinical Training for trans-nasal flexible laryngoscopy with Tiina Andersen: Tiina developed the testing procedure and will be key in teaching and developing my clinical skills. Two days will be spent observing, six days of hands-on teaching and skill development, and two days of competence sign off (10 days in total).
Clinical time at NMCCC, Queens Square: observing the adult trans-nasal fibreoptic laryngoscopy clinic (2 days) building knowledge and skills. In addition, discussions on developing ways of integrating a physiotherapy led trans-nasal fibreoptic laryngoscopy into clinical practice.
Clinical training for Opto-electronic plethysmography at Milan Polytechnic university with Antonella LoMauro: Building on an existing relationship, spending dedicated time in the OEP laboratory refining the methods of data collection and advancing methods of data analysis (4 days).

Building Collaborations

The international and industry collaborations I intend to establish are outlined below. These will support and enhance the proposal, as well as the career development programme, whilst also ensuring I develop partnerships for future research.
Haukeland University Hospital, Bergen, Norway: The second largest university hospital in Norway, which functions as a full-service public teaching hospital. The hospital provides significant training and research activities housing several specialty units that are internationally recognised as centres of excellence. Haukeland University Hospital endeavours to translate cutting edge research and teaching into improved patient outcomes and the best quality clinical care. A visit has been scheduled for the first year of my project and I am fortunate to Tiina Andersen as a practice supervisor who is based at Haukeland University Hospital. The 10 day visit will focus on training and collaboration to develop ongoing research development. The visit will focus on learning the technique for the project and working in the laboratory. It will reinforce links with European collaborations.
Collaborations from Industry: This project will link with different industry partners. This enables future potential for new opportunities, resources and building a network beyond the public sector. It will enhance the impact and relevance of the proposal:
ChilliPharm: This company will be providing the online video platform for video synchronisation. They are already well known in the medical field and have collaborated with allied health professionals in the past. They are leaders in utilising specialist videos for clinical trials and are happy to be collaborators on this proposal.
Breas: This company design and produce the mechanical insufflation devices. They are the leading company in the United Kingdom. They have worked alongside Great Ormond Street Hospital to provide devices to patients and are happy to continue to support this project. They are an international company and have excellent networking opportunities.
BTS Bioengineering and Milan Polytechnic university: The opto-electronic plethysmography equipment was provided by BTS bioengineering and supported by Milan Polytechnic University. They are both willing to continue to support the equipment throughout this project.
Storz: The trans-nasal fibreoptic laryngoscopes are managed by Storz. They provide on-site support at Great Ormond Street and are happy to collaborate on this project.

Conferences

CureSMA 2026: This international conference is a family centre conference and would be an excellent opportunity to discuss the study to continue PPIE on an international platform, prior to recruitment and testing.
International Conference on Home Mechanical Ventilation and European Respiratory Care Association Congress JIVDERCA 2028: This International multidisciplinary (MDT) conference will offer clinical learning and networking. The conference only meets every third year. There is a large, allied healthcare profession attendance. It will a good opportunity to present preliminary results to a respiratory network.
SMA Europe 2028: This conference is an excellent opportunity to disseminate study findings in the final year of PhD to the SMA network of clinicians and families.
World Muscle Society 2028: This International neuromuscular conference is an excellent platform to disseminate study findings in the final year of PhD to the neuromuscular network.
Other

Core Clinical Services Directorate

Date: Wednesday 7th May 2025

To Whom It May Concern

RE: Lisa Edel PHD application. Host organisation Letter of Support

It is with great pleasure that I write in support of Lisa Edel's PhD application. The partnership between the Physiotherapy Department at Great Ormond Street Hospital (GOSH) and the Neuromuscular and Respiratory section at the Institute of Child Health (ICH), University College London (UCL) provides an outstanding environment for Lisa to complete a doctoral fellowship and build a strong clinical academic career. A close relationship exists between the two departments and physiotherapists contribute significantly to UCL MSc programmes, providing specialist post-graduate lectures, offering clinical placements, and co-supervision of MSc dissertation projects.

At GOSH we aspire for research to be integral to the working lives of our staff and the patients and families that we treat. We learn from every patient, and using the knowledge gained improve the health of all patients at GOSH and children across the world. Research is considered a core component when recruiting to leadership positions across the organisation. All clinical divisions are supported to develop and own their research agenda and the Physiotherapy service is no exception; pro-active in both leading and collaborating on research, and development with other members of the multidisciplinary team at GOSH, and national and international centres.

We have on-going physiotherapy research projects (many with external funding) within all specialities in GOSH. As a department we support our staff to attend and present at national and international events, ensuring dissemination of research activity. We have experience of supporting others in pursuing a clinical academic career through the NIHR. As a result, we have built a questioning workforce with a commitment to drive change and improvement within and outside of GOSH for the benefit of patients, parents, staff and the wider NHS.

As a department, we have had five physiotherapists who have successfully defended their PhD and we currently have one physiotherapist who has been awarded the Clinical Doctoral Research Fellowship, I know they will all be able to provide additional support to Lisa during her fellowship and in her future clinical academic career.

The Physiotherapy department at GOSH are in full support of Lisa's application for this award; not only is this an excellent opportunity for her own development but it will also bring wider benefits to the Physiotherapy and the Neuromuscular Respiratory Services at GOSH, as well as to the broader Trust. She will continue to be embedded within the physiotherapy department contributing through regular clinical sessions, enabling her to maintain her high level of expertise in neuromuscular respiratory care, together with continuing to support and be a role model to colleagues involved in workplace research and audit activity.

As a GOSH clinician, Lisa will be able to access support from a variety of other sources. The Division of Research and Innovation sits alongside the Trust's clinical divisions and provides a one-stop shop for researchers, along with infrastructure to support research teams both in

developing and delivering research projects. Lisa will continue to obtain support from the Centre for Outcomes and Experience Research in Children's Health, Illness and Disability at GOSH. This dedicated research unit is led by Professor Faith Gibson, Deputy Chief Nurse for Research, who holds a joint appointment with the University of Surrey. The team comprises of Senior Research Fellows from differing allied health backgrounds and focuses on delivering research that seeks to improve the care, health and well-being of children and young people. Lisa would have access to the clinical academic careers programme which provides education and support specifically for nurses and allied health professionals. She will benefit from the experience of core members who have been Chief Investigator or co-applicants on NIHR/Wellcome/MRC funded projects, together with many other charity-funded projects. Outputs from the Centre during this time include peer-reviewed publications and conference presentations. They have also supported clinicians in securing funding at varying stages on the NIHR clinical academic pathway and would provide Lisa with on-going support beyond the fellowship.

I am delighted to support Lisa undertake a fellowship hosted by GOSH and partnered by UCL. This will further strengthen the department's relationship, ensuring clinically relevant research is completed which has a direct impact on patient care and healthcare service provision.

Yours sincerely,

Dr Tim Liversedge BSc(hons) MBChB FRCA MSc
Directorate Chief of Service and Consultant Anaesthetist
Great Ormond Street Hospital for Children NHS Foundation Trust
LONDON WC1N 1DZ

☎: 0207 405 9200
✉ GOSH email: tim.liversedge@gosh.nhs.uk

Great Ormond Street
Institute of Child Health
Developmental Neuroscience

19 May 2025

To whom it may concern

The partnership between the Physiotherapy Department at Great Ormond Street Hospital (GOSH) and the Neuromuscular and Respiratory section at the Institute of Child Health (ICH), University College London (UCL) will provide an outstanding environment for Lisa to complete a doctoral fellowship and build a strong clinical academic career. There is a close relationship between the two departments and physiotherapists contribute significantly to UCL MSc programmes, through providing specialist post-graduate lectures, offering clinical placements, and co-supervision of MSc dissertation projects. We confirm that, if the award is granted, Lisa will be integrated into the students at ICH enabling her to conduct research development activity appropriate to the fellowship. She will benefit from the experience of core members who have been Chief Investigator or co-applicants on NIHR/Wellcome/MRC funded projects, together with many other charity funded projects. She will also be supported by clinicians who have secured funding at varying stages on the NIHR clinical academic pathway and would provide Lisa with on-going support beyond the fellowship.

The Great Ormond Street Institute of Child Health (ICH), which is part of the Faculty of Population Health Sciences within the School of Life and Medical Sciences at UCL is exceptionally well equipped, both in terms of facilities and expertise to support this fellowship. ICH, together with its clinical partner GOSH, forms the largest concentration of children's health research in Europe. As a PhD student within the developmental neurosciences section Lisa would be integrated into a thriving academic community. The primary focus of the department is to minimise the impact of neurological disorders, improving prognosis and evaluating therapeutic strategies as well as optimising functional outcome. Lisa's project aligns with departments focus and she will benefit from being part of the Dubowitz Neuromuscular Centre, which provides a multidisciplinary service as a leading clinical and research centre. Within in this, she will have access to the clinical trials team for support, other PhD students as well as others who have continued the clinical academic pathway. ICH are enthusiastic in progressing students through clinical academia and Lisa will be sure to thrive in this environment.

UCL and ICH have excellent record of research with top class research facilities. In the Research Excellence Framework UCL was the top-rated university in the UK for research power (average research score multiplied by staff numbers) and also had the greatest 'impact power' of any university.

The department currently hosts PhD students and has a strong track record of supporting NIHR funded students. The facilities available at ICH/UCL and GOSH fully meet the range of requirements for hosting an NIHR Fellow.

Within UCL Lisa will have access to a network of 18 libraries with an excellent support network for research and academic writing. The section has vast experience of clinical physiotherapy and paediatric critical care research providing further teaching opportunities, interaction with fellow researchers working towards common goals, and guidance and mentoring from recognised physiotherapy, neuromuscular and respiratory experts. The question posed by this research is vital to on-going developments in respiratory management of children with spinal muscular atrophy and the patient's effectiveness of airway clearance and quality of life. Lisa has extensive

clinical experience in this area and is in an optimal position to complete this research, and as such I provide strong support for this application.

Supporting Lisa to undertake a fellowship hosted by GOSH and partnered by UCL will further strengthen the department's relationship, thus ensuring clinically relevant research is completed which has a direct impact on patient care and healthcare service provision.

Sincerely,

Torsten Baldeweg, MD

Professor of Cognitive Neuroscience
Head of Department

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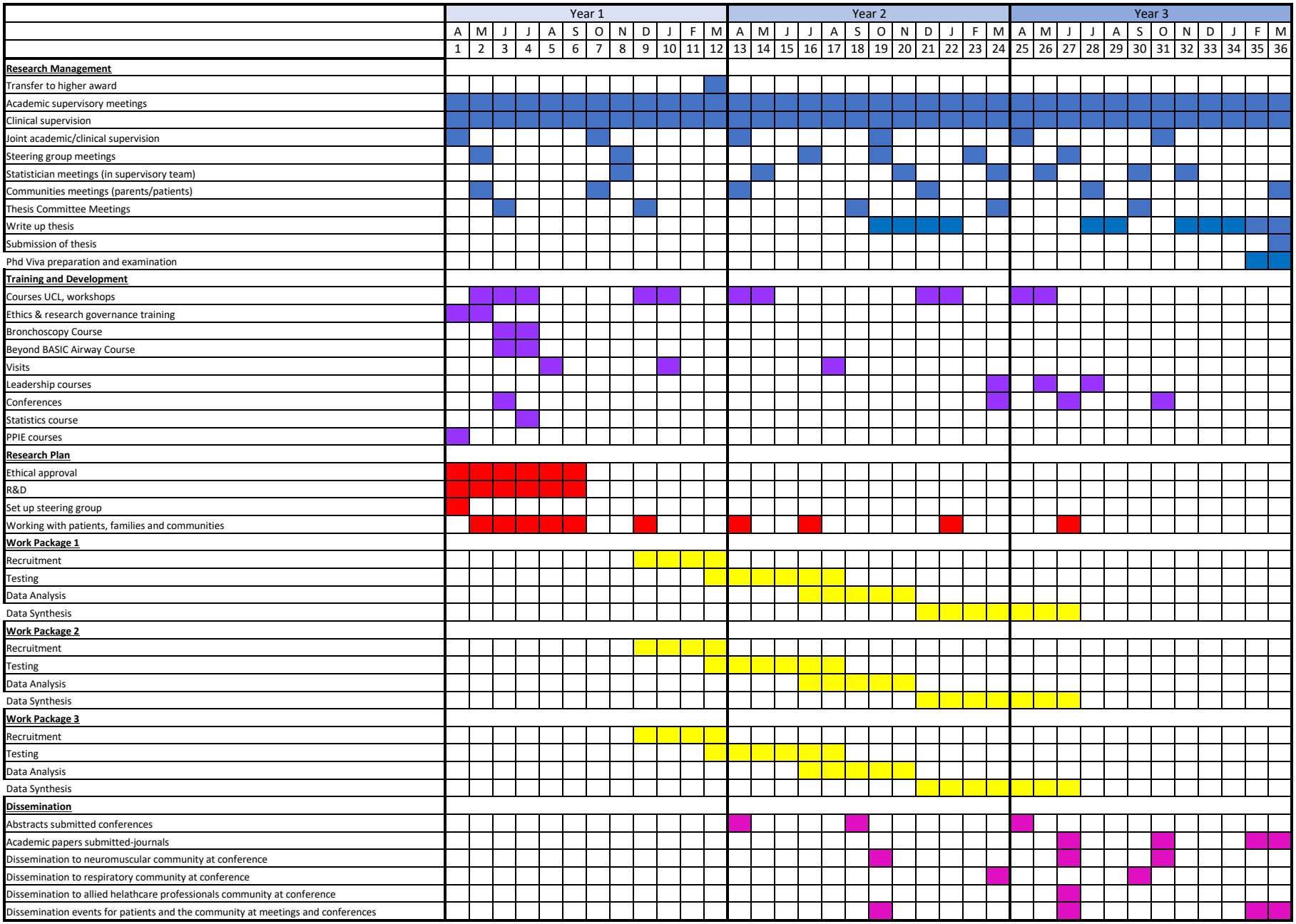
Year 1
Year 2
Year 3
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Research Management
Transfer to higher award
Academic supervisory meetings
Clinical supervision
Joint academic/clinical supervision
Steering group meetings
Statistician meetings (in supervisory team)
Communities meetings (parents/patients)
Thesis Committee Meetings
Write up thesis
Submission of thesis
Phd Viva preparation and examination
Training and Development
Courses UCL, workshops
Ethics & research governance training
Bronchoscopy Course
Beyond BASIC Airway Course
Visits
Leadership courses
Conferences
Statistics course
PPIE courses
Research Plan
Ethical approval
R&D
Set up steering group
Working with patients, families and communities
Work Package 1
Recruitment
Testing
Data Analysis
Data Synthesis
Work Package 2
Recruitment
Testing
Data Analysis
Data Synthesis
Work Package 3
Recruitment
Testing
Data Analysis
Data Synthesis
Dissemination
Abstracts submitted conferences
Academic papers submitted-journals
Dissemination to neuromuscular community at conference
Dissemination to respiratory community at conference
Dissemination to allied helathcare professionals community at conference
Dissemination events for patients and the community at meetings and conferences

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